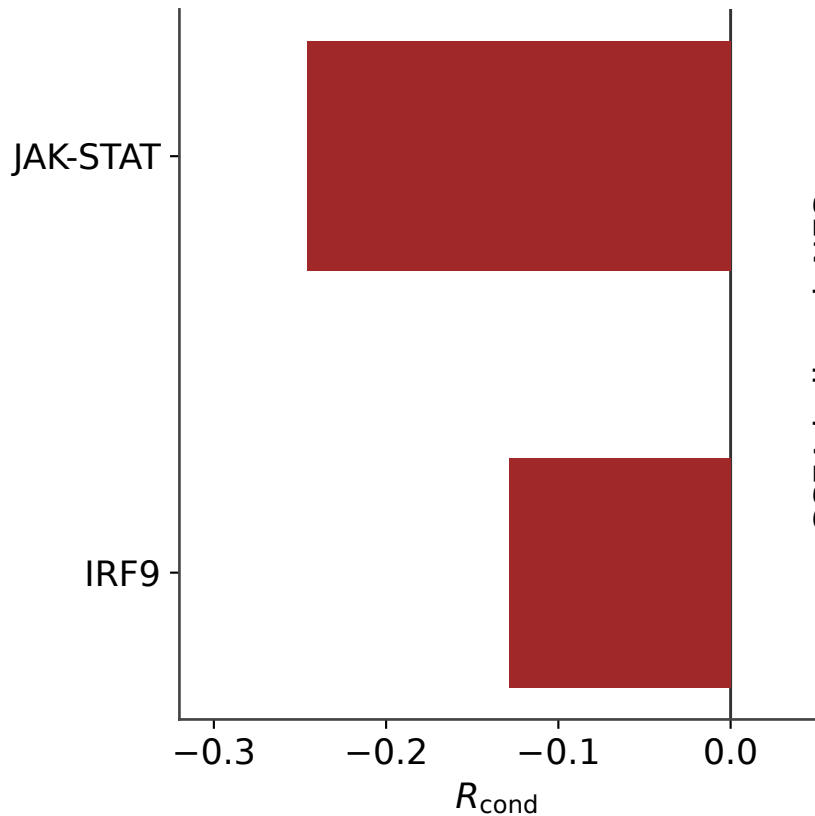
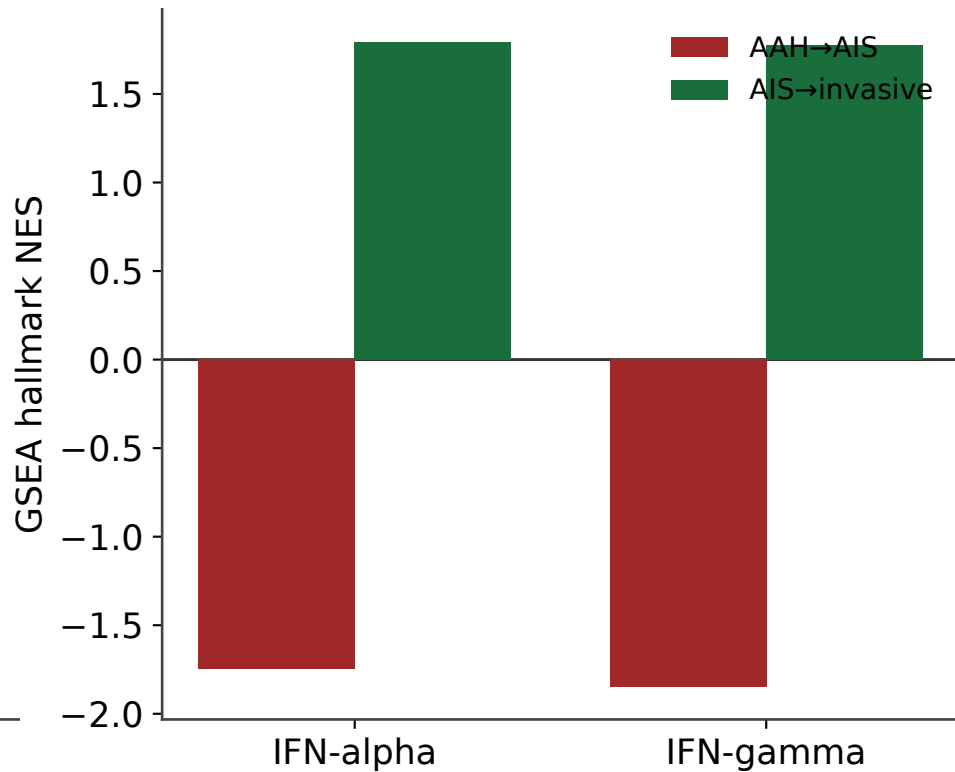


Context residual (AAH→AIS)



Whole-transcriptome enrichment



two readouts with different confounds agree on the preinvasive loss (all $p_{\text{adj}} < 10^{-3}$)